Description

Image





Caption

1. Car rear light casing. © Chris Lefteri 2. PMMA chair. © Chris Lefteri

The material

When you think of PMMA, think transparency. Acrylic, or PMMA, is the thermoplastic that most closely resembles glass in transparency and resistance to weathering. The material has a long history: discovered in 1872, first commercialized in 1933, its first major application was as cockpit canopies for fighter aircraft during the second World War.

Compositional summary

(CH2-C(CH3)COOCH3)n

General properties

Density	1.17e3	-	1.2e3	kg/m^3
Price	* 1.43	-	1.76	GBP/kg
Date first used	1933			

Mechanical properties

Young's modulus	2.24	-	3.24	GPa
Shear modulus	* 0.803	-	1.16	GPa
Bulk modulus	* 4.18	-	4.39	GPa
Poisson's ratio	* 0.387	-	0.403	
Yield strength (elastic limit)	54	-	72	MPa
Tensile strength	54	-	72	MPa
Compressive strength	* 72.4	-	124	MPa
Elongation	2	-	5.5	% strain
Hardness - Vickers	* 16	-	22	HV

Fatigue strength at 10^7 cycles		15.2	-	16.8	MPa
Fracture toughness		0.7	-	1.6	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.0123	-	0.0179	

Thermal properties

Glass temperature		99.9	-	110	°C
Maximum service temperature		43.9	-	55.9	°C
Minimum service temperature		-75.2	-	-65.2	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity		0.167	-	0.251	W/m.°C
Specific heat capacity		1.4e3	-	1.52e3	J/kg.°C
Thermal expansion coefficient		90	-	162	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity		3.3e23	-	3e24	µohm.cm
Dielectric constant (relative permittivity)		3.2	-	3.4	
Dissipation factor (dielectric loss tangent)		0.05	-	0.06	
Dielectric strength (dielectric breakdown)		15.7	-	19.7	MV/m

Optical properties

Transparency		Optical Quality			
Refractive index		1.49	-	1.5	

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		3	-	5	
Moldability		4	-	5	
Machinability		3	-	4	
Weldability		5			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Excellent			
Soils, alkaline (clay)		Excellent			
Wine		Excellent			

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Unacceptable
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Excellent
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Excellent
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Limited use
Phosphoric acid (85%)	Unacceptable
Sulfuric acid (10%)	Unacceptable
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Unacceptable
Benzene	Unacceptable
Carbon tetrachloride	Limited use
Chloroform	Unacceptable
Crude oil	Acceptable
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Acceptable
Petrol (gasoline)	Excellent
Silicone fluids	Limited use
Toluene	Unacceptable
Turpentine	Unacceptable
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Unacceptable
Acetone	Unacceptable
Ethyl alcohol (ethanol)	Limited use
Ethylene glycol	

	Limited use
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Unacceptable

Durability: halogens and gases

Chlorine gas (dry)	Limited use
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Good

Durability: flammability

Flammability	Highly flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 6.17	-	6.81	kg/kg
Embodied energy (virgin grade production)	* 118	-	131	MJ/kg
Water usage	* 72.3	-	79.9	l/kg

Material processing: energy

Polymer extrusion, embodied energy	* 8.03	-	8.88	MJ/kg
Polymer molding (injection, blow), embodied energy	* 20	-	23	MJ/kg
Machining, coarse, embodied energy	* 18.2	-	22.8	MJ/kg
Machining, fine, embodied energy	* 25.4	-	36.1	MJ/kg
Machining, grinding, embodied energy	* 32.7	-	49.4	MJ/kg

Material processing: climate change

Polymer extrusion, climate change (CO2-eq)	* 0.527	-	0.585	kg/kg
Polymer molding (injection, blow), climate change (CO2-eq)	* 1.38	-	1.6	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.19	-	1.49	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.66	-	2.36	kg/kg
Machining, grinding, climate change (CO2-eq)	* 2.14	-	3.24	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 2.04	-	2.25	kg/kg
Embodied energy, recycling	* 22.4	-	24.9	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 25.9	-	27.2	MJ/kg
Combustion CO2	* 2.15	-	2.25	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Acrylics are non-toxic and recyclable.

Recycle mark



Supporting information

Design guidelines

Acrylic, PMMA, is hard and stiff as polymers go, easy to polish but sensitive to stress concentrations. It shares with glass a certain fragility, something that can be overcome by blending with acrylic rubber to give a high-impact alloy (HIPMMA). PVC can be blended with PMMA to give tough, durable sheets. Acrylic is available as a sheet, rod or tube and can be shaped by casting or extrusion. Cell casting uses plates of glass and gasketing for a mold; it allows clear and colored panels up to 4 inches thick to be cast. Extrusion pushes melted polymer pellets through a die to give a wide variety of shapes, up to 0.25 inches thick for sheet. Clear and colored PMMA sheet lends itself to thermoforming, allowing inexpensive processing. A hybrid sheet manufacturing process, continuous casting, combines the physical benefits of cell casting and the cost efficiency of extrusion. Extruded and continuous cast sheet have better thickness tolerance than cell-cast sheet. PMMA can be joined with epoxy, alpha-cyanoacrylate, polyester or nitrile-phenolic adhesives. It scratches much more easily than glass, but this can be partially overcome with coatings.

Technical notes

Polymers are truly transparent only if they are completely amorphous - that is, non-crystalline. The lumpy shape of the PMMA molecule ensures an amorphous structure, and its stability gives good weathering resistance. PMMA is attacked by esters, ketones, acids and hydrocarbons, and has poor resistance to strong acids or bases, solvents and acetone.

Typical uses

Lenses of all types, cockpit canopies and aircraft windows, signs, domestic baths, packaging, containers, electrical components, drafting equipment, tool handles, safety spectacles, lighting, automotive tail lights, chairs, contact lenses, windows, advertising signs, static dissipation products, compact disks.

Tradenames

Acrive, Acrylite, Acryrex, Altuglas, Cyrolite, Diakon, Glasflex, Goldrex, Lucite, Lucryl, Optix, Oroglas, Perspex, Plexiglas, Plexit, Sumiplex

Links

ProcessUniverse

Producers

Reference